Specification for Z-Test Statistical Framework

HYPOTHESIS TESTING STANDARDS & VALIDATION PROTOCOL (V2)

This technical specification defines the mandatory mathematical architecture, structural constraints, and validation thresholds for executing a **Z-Test**. This standard governs applications comparing a sample group mean against a known population benchmark or comparing two distinct sample sets under conditions of known variance.

1. System Prerequisites & Assumptions

Before a Z-test formula is executed within an analysis engine, data pipelines must programmatically validate the following four prerequisite conditions:

- **Known Population Variance:** The population standard deviation (σ) must be an explicitly specified, known scalar value. If variance is unknown, execution must default to a Student's t-test pipeline.
- **Sample Volume Threshold:** The central limit theorem requires a large sample size—conventionally defined as $n \geq 30$—to ensure normal distribution of the sample mean.
- **Independence:** Data records must be collected independently. Sampling must be executed with replacement, or represent less than 10% of the aggregate population.
- **Normality:** The raw underlying data must follow a normal distribution or adhere to the central limit theorem threshold specified above.

2. Mathematical Formulations

2.1 One-Sample Z-Test

Used when validating whether an experimental sample mean deviates significantly from an established, baseline population mean.

$$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

Where: $\bar{x}$ = Sample Mean, μ = Population Mean, σ = Population Standard Deviation, and n = Sample Size.

3. Decision Rule Matrix & Standard Thresholds

The standard critical value thresholds for evaluating the computed Z score against a standard normal distribution ($N(0, 1)$) are defined below:

Confidence Level	Alpha (α) Significance	Two-Tailed Critical Z-Score ($|Z|$)	One-Tailed Critical Z-Score (Z)
90%	0.10	`± 1.645`	`1.28`
95%	0.05	`± 1.960`	`1.645`
99%	0.01	`± 2.576`	`2.33`

Evaluation Logic:

If $|Z_{computed}| \geq Z_{critical}$, reject the Null Hypothesis (H_0) and declare statistical significance. If the calculated value is lower, fail to reject H_0.

4. Reference Execution Example with Raw Dataset & Results

This section serves as a programmatic test case to verify compiler and calculator logic compatibility.

Scenario: A database cluster optimization pass is implemented to improve transactional query throughput. The engineering team must confirm if the optimization pass shifts performance benchmarks away from historical baselines.

4.1 Known Population Parameters (Historical Baseline)

- Historical Population Mean (μ): **500.00 queries/sec**
- Known Population Standard Deviation (σ): **40.00 queries/sec**

4.2 Collected Sample Experimental Data (n = 40)

The execution run captures the following 40 discrete throughput observation points during a performance-testing window:

```
502.4, 510.1, 498.5, 521.3, 508.9, 515.2, 489.6, 530.4, 512.2, 505.7,
518.3, 495.1, 522.6, 509.4, 511.0, 528.1, 503.7, 514.8, 499.2, 516.5,
525.0, 507.3, 513.9, 492.4, 519.8, 506.1, 524.3, 511.6, 501.9, 527.2,
510.5, 497.8, 520.1, 513.0, 509.2, 517.4, 522.0, 504.6, 516.1, 514.5
```

4.3 Verification Computations

Step 1: Compute Sample Metrics

Sum of observations (Σx) = `20,443.10`

Sample Size (n) = `40`

Calculated Sample Mean ($\bar{x}$) = $20{,}443.10 / 40$ = **511.0775 queries/sec**

Step 2: Determine Standard Error of the Mean (SE)

$SE = \sigma / \sqrt{n} = 40.00 / \sqrt{40} = 40.00 / 6.324555 = 6.324555$

Step 3: Calculate the Z-Test Statistic

$Z = (\bar{x} - \mu) / SE = (511.0775 - 500.00) / 6.324555 = 11.0775 / 6.324555 = 1.7515$

4.4 Compiled Statistical Results

Execution Metric	Calculated Output Value	Structural Rule Mapping / Bounds
Hypotheses Definition	$H_0: \mu = 500$; $H_a: \mu \neq 500$	Two-tailed structural configuration framework.
Computed Z-Score	`1.752`	Evaluated against standard normal criteria.
Critical Value (α = 0.05)	`± 1.960`	Rejection boundaries for 95% Confidence Level.
P-Value (Two-Tailed)	**0.0798**	Probability of observing extreme deviation by chance alone.
Operational Status	**FAIL TO REJECT H_0**	Since $\lvert 1.752 \rvert < 1.960$ (and $p > 0.05$).

4.5 Analytical Verdict

The computed test metric $Z = 1.752$ falls within the non-rejection region bounded by ± 1.960. Therefore, at the $\alpha = 0.05$ significance level, the engine **fails to reject the Null Hypothesis**. While the raw sample mean shows an apparent increase to 511.08 queries/sec, the difference is statistically insufficient to prove that the optimization pass altered the underlying performance baseline rather than random performance variation.